Involuntary responses 6a

1 This information about involuntary responses has been mixed up. Read through the information and arrange the statements in the correct order by numbering them.

a a signal is sent along sensory neurones to the spinal cord.	b are asleep. Reflexes occur the following way: A stimulus
c aware of the movements that we are making	d by the brain. We do not usually become
e effectors. Consequently the reflex action is produced.	f Examples of reflexes include the fast eye blink when a fly
g in a reflex action. Reflexes can happen even while we	h In the spinal cord connector neurones switch in the normal
i Involuntary responses are called reflexes. Involuntary	j is detected by receptors in a sense organ and as a result
k is just about to fly into your eye, and the mouth watering	l or not. They are a very fast kind of
m response and then pass the message to the appropriate	n response, since they do not involve any decision making
o responses happen whether we want them to	p when you see or smell a yummy piece of food.

2 Cut out the sentence parts, arrange them neatly into three paragraphs and glue them onto a sheet of paper. You may need to cut some of the sentence fragments in half to make logical paragraphs.

3 Briefly write the function of each paragraph.

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Looking at language

4 Verbs are central to a sentence because they tell us what is going on, what action is taking place. Verbs often show how a participant relates to something else. There are different kinds of verbs:

- Linking verbs connect things with their characteristics. The most common linking verbs are *are* and *have*. Often a linking verb can be replaced with a form of *are* or *have*.
- Action verbs describe the action that is occurring in the clause, for example *run* and *explode*.
- Saying verbs show what somebody or something is saying, for example *shout* and *show*.
- Thinking verbs describe what somebody is thinking or feeling, for example *conclude* and *decide*. Saying and thinking verbs are quite rare in scientific writing.

a In your ordered paragraphs above, underline with a straight line any action verbs, underline with a dotted line any linking verbs, circle any (saying) verbs and underline with a wavy line any thinking verbs. Note that verbs can be more than one word long.

b State which verb type dominates each paragraph. _____

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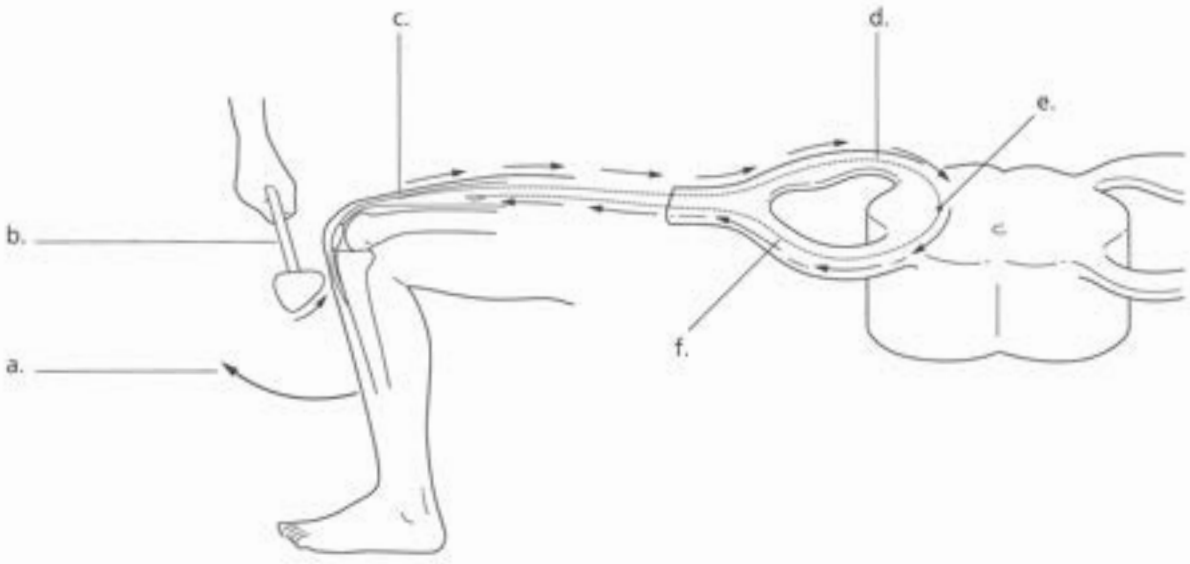
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Involuntary responses 6b

5 Cut out each square below and use them to label the diagram about the knee jerk reflex.

A RESPONSE: the knee straightens as leg muscles contract.	B Sensory neurons carry the signal from the RECEPTOR.	C Motor neurons carry the message to the EFFECTOR.	D Stretch RECEPTORS in the leg muscle fire off a signal.	E Connector neurones connect to motor neurones in the spinal cord.	F STIMULUS: a tap stretches the tendon below the knee.



6 Order the events described in the squares above in their correct order, starting with the stimulus.

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